

1 **Online Methods**

2 **Datasets and registration**

3 Breast Atera WTA was used as the discovery dataset and cervical Atera WTA as a second
4 epithelial cancer stress test. Each dataset was analyzed as paired spatial whole-transcriptome
5 (WTA) and H&E whole-slide image data. Atera WTA coordinate metadata in the SpatialData
6 object were used to register H&E-derived image features and WTA measurements into a shared
7 contour coordinate system. The analysis did not redistribute raw 10x input files; figure source-
8 data tables and machine-readable supplementary data tables are provided with the submission
9 package.

10 **Contour origin**

11 Spatial-omics contours were generated from Atera WTA spatial-transcriptomics inputs,
12 including cell-coordinate tables and cluster or cell-group definitions. The contour-generation
13 workflow supports contour synthesis from spatial cell coordinates, clustered cell labels and
14 spatial-transcriptomics contour exports. Separate H&E image-segmentation functionality was not
15 used to draw the breast S1-S5 contours or the cervical tissue contours in this study. H&E images
16 were not used to draw contours. H&E images entered only after contour generation, through
17 assignment of LazySlide image features to the registered contour geometry. This contour origin
18 makes the test conservative: the contour label already captures coarse molecular-spatial tissue
19 structure, and mTM asks whether H&E morphology contributes residual WTA information after
20 that structure has been controlled.

21 **H&E WSI embedding**

22 The original H&E OME-TIFF was converted into a tiled pyramidal BigTIFF readable by the
23 direct WSI workflow. LazySlide was run on the converted WSI with PLIP for the primary

analysis and UNI for model-sensitivity analysis where available. Tile-level image embeddings were assigned to spatial-omics contours using registered spatial geometry, and contour-level image summaries were computed from the tiles assigned to each polygon. No H&E-derived contour segmentation was introduced during this step. The direct WSI, contour aggregation and mTM workflow is described here; implementation details are available to editors through the editorial submission materials.

WTA program scoring

Atera WTA measurements were summarized within the same contour polygons used for image-feature aggregation. Program scores were assembled for biologically interpretable families including endocrine/epithelial identity, metabolic/stress, stromal-remodeling/CAF/ECM, immune ecology/TLS/immune exclusion and invasion/boundary/EMT. Breast analyses focused on luminal estrogen response, unfolded protein response and oxidative phosphorylation in S3 contours. Cervical analyses tested stromal, immune, invasion, epithelial and stress/metabolic programs as a cross-cancer stress test. Program scores were treated as continuous contour-level WTA readouts rather than as categorical labels.

For component-gene audit analyses, contour-level normalized means were recomputed for the genes comprising each selected breast S3 WTA program. The reported H&E embedding axis was then tested against each component gene using the same rank-residualized partial Spearman framework and the same spatial covariates. These analyses were used as transcript-level sanity checks only; they are not independent IHC or protein validation.

Contour-level aggregation

For contour i , let T_i be the set of LazySlide tiles assigned to that contour and G_i the cells or transcripts assigned to the same contour polygon. H&E image features were summarized as

contour-level means or related summaries across T_i , and WTA program scores were summarized across G_i . These paired summaries define a contour-level table in which each row has the same geometry, the same spatial-omics-derived contour label and matched H&E/WTa measurements.

Non-independence of contour construction and residual estimand

The spatial-omics contours are not independent histological labels. We therefore define the estimand as a within-label residual association rather than an independent label-prediction task. For contour i , let C_i denote the spatial-omics-derived discrete contour label, S_i denote spatial covariates including centroid coordinates and boundary-distance bins, X_{ik} denote H&E foundation-model feature k , and Y_i denote a continuous WTA program score. The covariate design matrix is:

$$Z_i = [1, \text{onehot}(C_i), S_i].$$

For breast S3 analyses, C_i is constant by design, so the contour-label one-hot term is omitted.

Rank-transformed image and WTA variables are residualized against Z :

$$r_k^X = (I - P_Z) \text{rank}(X_k), r^Y = (I - P_Z) \text{rank}(Y),$$

where $P_Z = Z(Z^T Z)^{-1} Z^T$ is the projection matrix. The reported residual association is

$$\rho_k = \text{cor}(r_k^X, r^Y),$$

computed as a Spearman correlation after rank residualization. This design cannot establish standalone morphology-only prediction and does not exclude every form of contour-construction-induced coupling, but it directly tests whether H&E features map continuous WTA program variation beyond simply rediscovering the discrete spatial-omics contour label.

Residual decoding and program selection

For each candidate WTA program, image features were screened by absolute residual association within the relevant contour set. The primary breast results were restricted to S3 contours and

reported the strongest luminal estrogen-response, unfolded-protein-response and oxidative-phosphorylation associations. Cervical analyses used the same residual-decoding logic but interpreted positive findings as a cross-cancer stress test rather than direct replication of the breast luminal phenotype. Cervical claims are based on embedding-derived WTA program associations; PLIP text-similarity outputs from the breast-oriented prompt set were not used as cervical biological claims. Individual PLIP or UNI embedding dimensions were not assigned universal biological meaning because latent axes can rotate, flip or change scale across foundation models.

Spatial permutation and block bootstrap

For the spatial-null defense, molecular residuals were permuted within strata preserving contour label, centroid-position bins and boundary-distance bins. This retains coarse compartmental and spatial organization while breaking contour-wise H&E-to-WTA pairing. For B permutations, empirical two-sided enrichment was calculated as:

$$P_{emp} = \frac{1 + \sum_{b=1}^B 1(|\rho_b^{null}| \geq |\rho_{obs}|)}{1 + B}.$$

The permutation mitigates coarse spatial-autocorrelation explanations but does not exclude all fine-scale spatial dependence, registration uncertainty or biological coupling induced by the original spatial-omics contour construction. Spatial block bootstrap resampled centroid x/y spatial blocks and recomputed ρ to estimate 95% confidence intervals. Additional sensitivity checks recomputed candidate associations after leaving out each spatial block, compared observed pairings with local mismatched-pair controls and jittered centroid covariates by up to 1% of slide span.

For registration perturbation, existing tile-level embeddings were reassigned to contour polygons after 23 small coordinate perturbations: x/y translations of 8-32 pixels, rotations of 0.25-0.5 degrees, 0.5% scale changes and mild combined shifts. Candidate contour-level embedding means were recomputed after each reassignment and tested against the same WTA program values and covariates. This test probes whether the reported associations are brittle to small tile-to-contour registration changes without rerunning the H&E encoder.

For nested spatial holdout, contours were partitioned into a 4 by 4 centroid grid. In each fold, the embedding dimension was selected using only the training spatial blocks, and the selected feature and the locked manuscript feature were then evaluated on the held-out block. This stricter check is reported as fold-level sign stability, not as a new discovery screen, because small held-out blocks can be noisy for weaker cross-cancer/model stress-test candidates.

Boundary co-variation

Boundary profiles were computed in distance rings around selected tissue interfaces. For each selected program-feature pair, WTA program values and oriented H&E feature values were summarized across signed distance bins from contour boundaries. Candidate molecularly active zones were described only as ring-level H&E-WTA co-variation. No causal, temporal or directional boundary interpretation is claimed.

Software and reproducibility

The mTM analysis, direct WSI embedding, contour aggregation, statistical defense, figure composition and submission-package preparation are implemented in code available to editors through the editorial submission materials. Reviewer-facing source-data tables, supplementary data tables and robustness summaries are included in the anonymized submission package.

Data availability

Figure source-data CSVs are provided for Fig. 1 panels and Supplementary Figs. 1-3. Separate machine-readable Supplementary Data CSVs are provided for spatial-sensitivity summaries, component-gene audits, registration-perturbation tests and nested spatial holdout tests. Raw 10x Genomics Atera WTA and H&E input files are not redistributed by this manuscript package; the breast and cervical source datasets are available from 10x Genomics at <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-breast-cancer> and <https://www.10xgenomics.com/datasets/atera-wta-ffpe-human-cervical-cancer>. No IHC or protein-validation data were generated for this study.

Code availability

The mTM analysis, direct WSI embedding, contour aggregation, statistical defense, figure composition and submission-package preparation code is available in the pyXenium repository at <https://github.com/hutaobo/pyXenium>. The fixed software release used for this submission is pyXenium v0.4.6 at <https://github.com/hutaobo/pyXenium/releases/tag/v0.4.6>, and the submission-packaging hardening commit is 3c4617010bb1c76e18921958c4507f91512aa9dc at <https://github.com/hutaobo/pyXenium/commit/3c4617010bb1c76e18921958c4507f91512aa9dc>. Manuscript-level replication scripts are provided under manuscript/mtm_wta_nbt_replication/. The exact journal-upload package and individual submission files are archived at <https://github.com/hutaobo/pyXenium/releases/tag/nbt-submission-upload-20260518-draft>. HistoSeg contour-generation software is maintained at <https://github.com/hutaobo/HistoSeg>.

133 **Large language model assistance**

134 OpenAI Codex was used to assist with manuscript-package formatting, editorial consistency
135 checks and automated QA scripting. The author reviewed the final text, analyses and claims, and
136 takes responsibility for the submitted work.